

PREM YELWANDE

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Professional Summary

AI Engineer specializing in the design, architecture, and deployment of intelligent backend systems and production-ready AI applications. Extensive expertise in building LLM-powered multi-agent systems, orchestrating multi-agent pipelines, and integrating LLMs into scalable, production-grade backend architectures. Skilled in end-to-end system design from agent orchestration and tool/function integration to robust, high-performance Python-based infrastructure with a strong focus on production reliability and scalability.

Skills

- **Languages:** Python, Java, SQL
- **Backend:** FastAPI, REST APIs, SQLAlchemy, Pydantic, JWT/OAuth2, Async/Await, Middleware, Dependency Injection, Pytest, CORS, Caching, Rate Limiting
- **AI Stack:** LangChain, LangGraph, RAG, Tool/Function Calling, Prompt Engineering, LLM Evaluation, Guardrails, MCP
- **AI/ML:** LLMs, AI Agents, Multi-Agent Systems, Embeddings, Vector Search, Fine-tuning, NLP, Computer Vision
- **Databases:** PostgreSQL, MySQL, Redis, ChromaDB, PGVector, Schema Design, Query Optimization
- **Frameworks & Libraries:** TensorFlow, Keras, Scikit-learn, XGBoost, OpenCV, Pandas, NumPy, Matplotlib
- **AI Tools & APIs:** Hugging Face, Whisper, Tavily, Playwright, Mistral AI, OpenAI API, Gemini API, Sarvam AI, BeautifulSoup
- **DevOps & Cloud:** Git/GitHub, Docker, CI/CD (GitHub Actions), AWS Fundamentals
- **Coursework:** OOP, DBMS, OS, Computer Networks, Maths & Statistics

Projects

Autonomous AI Software Engineer Aug 2026 – Present

- Engineered an Autonomous AI software engineering system capable of independently planning, architecting,generating code.
- Designed a multi-agent LangGraph pipeline with specialized planner, architect,coder agents for end-to-end task execution.
- Developed a FastAPI backend paired with a custom dark glassmorphism frontend for real-time interaction and monitoring.
- **Live Link :** <https://devin-clone-oe17.onrender.com/>
- **Tech Stack :** Python,LangChain, LangGraph, FastAPI, Gemini , Multi-Agent Systems

Multi-Agent AI Research System Aug 2026 – Present

- Engineered a multi-agent autonomous AI system for automated web research, retrieval, extraction, and report generation
- Implemented specialized agents for web search, webpage reading and scraping, research synthesis, and content evaluation.
- Integrated LLMs with external tools and APIs to retrieve, process, and synthesize information from diverse web sources.
- **Live Link :** <https://prem-yelwande-multi-agent-system.streamlit.app/>
- **Tech Stack :** Python, LangChain, LangGraph, Mistral AI, Tavily, BeautifulSoup

Mental Health Score Predictor Aug 2026 – Present

- Developed a machine learning model to predict mental health scores from a dataset of 5,000 records and 12 input features.
- Built a lightweight FastAPI backend with Pydantic validation to accept user inputs and serve real-time model predictions.
- Achieved 87 percent model accuracy and successfully deployed the full application with a clean, interactive web interface
- **Live Link :** <https://mental-health-scorer-1-idnr.onrender.com/>
- **Tech Stack :** Python, Pandas, Scikit-learn, Random Forest, FastAPI, Pydantic

Role of Responsibility

Co-Head of Events Information Technology Students Association (ITSA), VIT Feb 2025 – Feb 2026

Planned and coordinated technical events and student activities under ITSA.
Coordinated event operations, student teams, and participants to ensure smooth execution.
Contributed to the planning and execution of SURGE, a five-day flagship technical event.

<https://vit.edu.in/itsa/>

Education

Vidyalankar Institute of Technology (VIT), Mumbai 2024 – 2028

B.Tech in Information Technology

Jai Hind College, Mumbai 2022 - 2024

Higher Secondary Certificate (HSC)

Certifications

Python for Web Development CDAC | June 2026

Java with DSA Udemy | August 2026